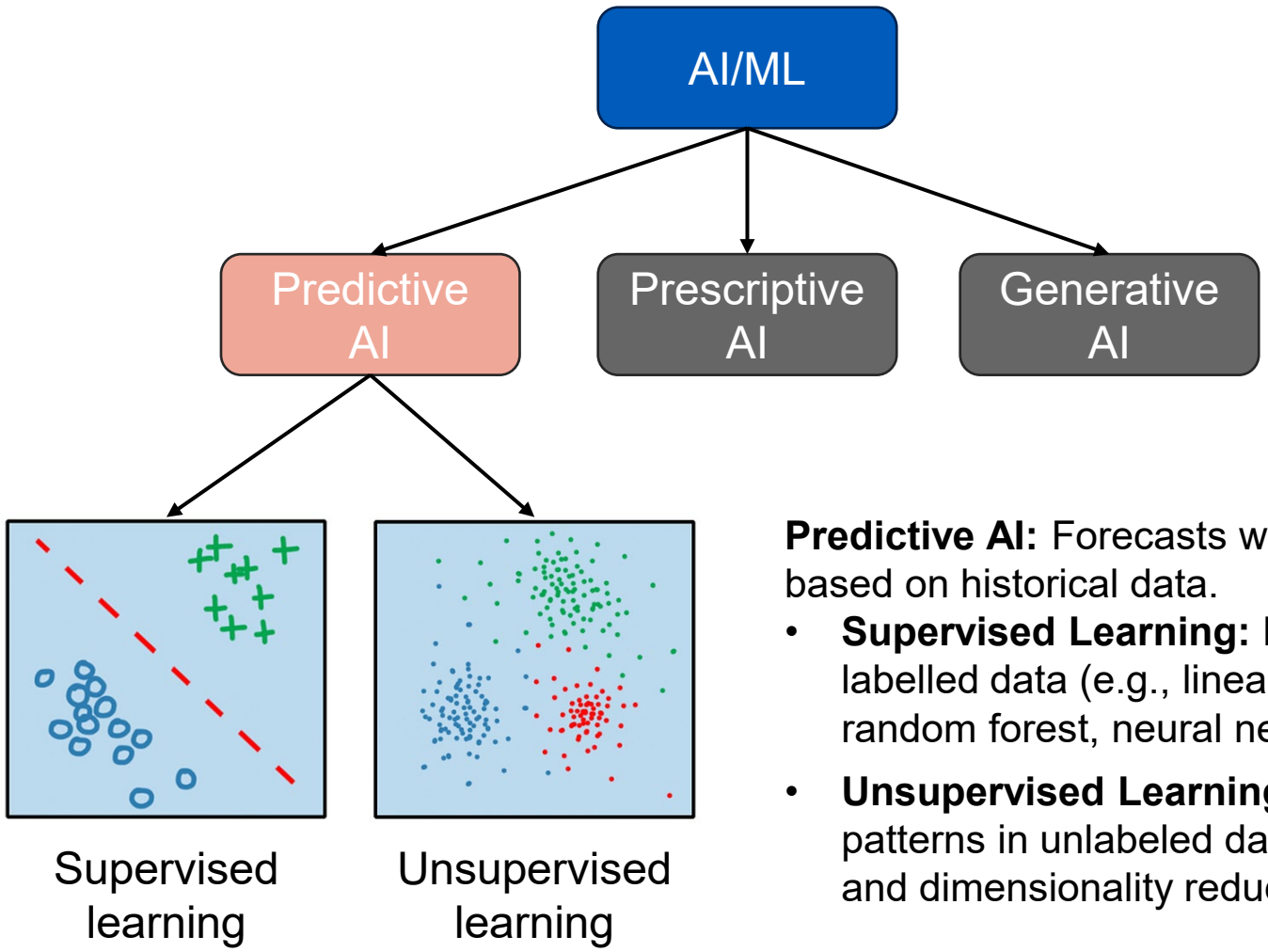
CE 400/500

Special Topics: AI for Chemical Engineers

Lecture 15: Introduction to Unsupervised Learning

University at Buffalo
The State University of New York

Predictive AI: Supervised and Unsupervised Learning



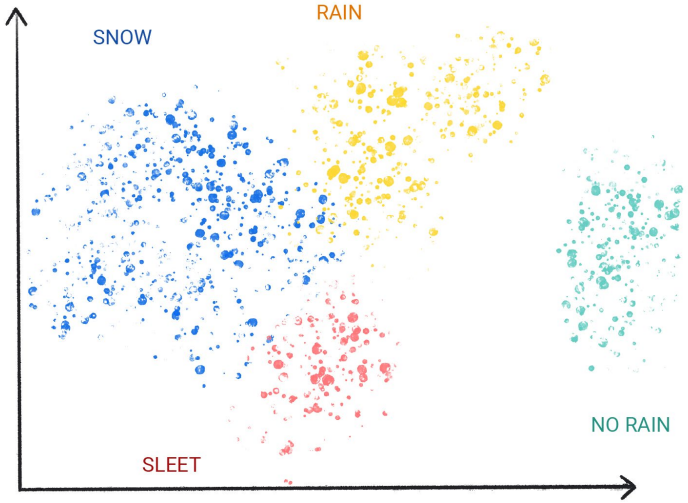
Predictive AI: Forecasts what will happen based on historical data.

- **Supervised Learning:** Learning from labelled data (e.g., linear regression, random forest, neural network).
- **Unsupervised Learning:** Discovering patterns in unlabeled data: clustering and dimensionality reduction.

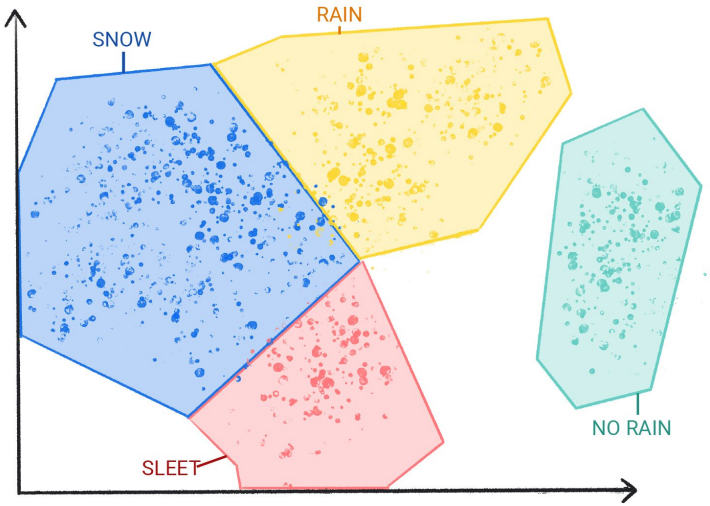
Unsupervised Learning: Clustering

Unsupervised learning models make predictions by finding patterns in a dataset, typically an unlabeled dataset.

The most common use of unsupervised machine learning is to cluster data into groups of similar examples:



An ML model clustering similar weather patterns.

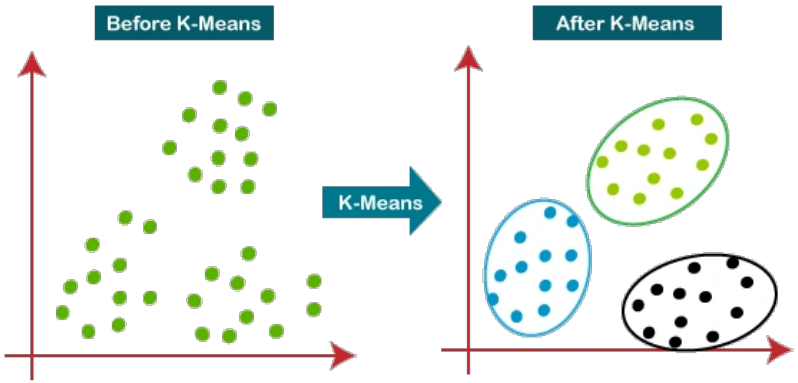


Clusters of weather patterns labeled as snow, sleet, rain, and no rain.

Clustering can help when useful labels are scarce or absent (e.g., anti-fraud, reaction or sensor data in extreme conditions in industry).

https://developers.google.com/machine-learning/intro-to-ml/what-is-ml

K-Means Clustering



Credit: Analytics Vidhya

K-Means aims to partition data into k clusters such that each data point belongs to the cluster with the nearest mean (called the **centroid**).The algorithm **iteratively** minimizes the within-cluster sum of squares by alternating between two steps:

Step A. Assign each data point $\mathbf{x}_i$ to the cluster whose centroid $\boldsymbol{\mu}_k$ is the closest to minimize the loss:

$$L = \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2$$

Step B. Recalculate each centroid as the mean of all points assigned to that cluster:

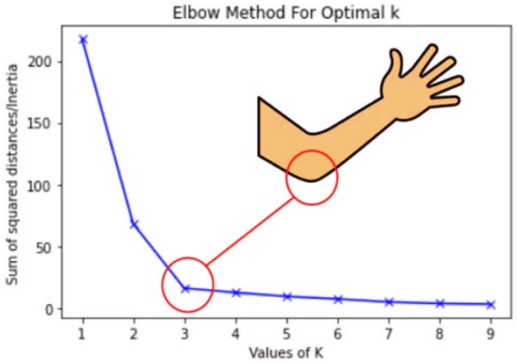
$$\boldsymbol{\mu}_i = \frac{1}{N_i} \sum_{\mathbf{x} \in S_i} \mathbf{x}$$

This k-means algorithm is illustrated in this online demo (made with Google AI Studio).

K-Means Clustering: Choosing k

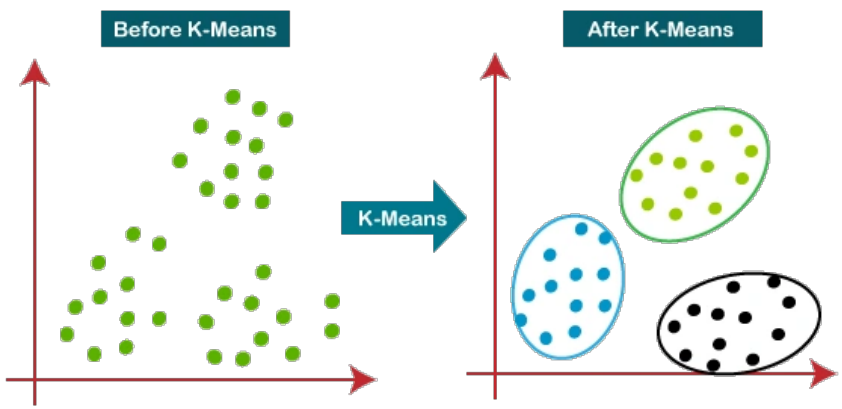
1. **Manual Choice of k**
   - Used when domain knowledge or prior expectation suggests a reasonable number of clusters.
   - Common in exploratory analysis or when a fixed number of groups is desired.
   - Enables easy control and interpretation but may overlook natural group structure.

2. **Automatic Choice of k:**
   - Applied when the true number of clusters is unknown.
   - Uses quantitative criteria to evaluate multiple k values: e.g., Elbow method: look for the point where improvement in loss slows down.



Line plot between K and inertia

K-Means Clustering: History



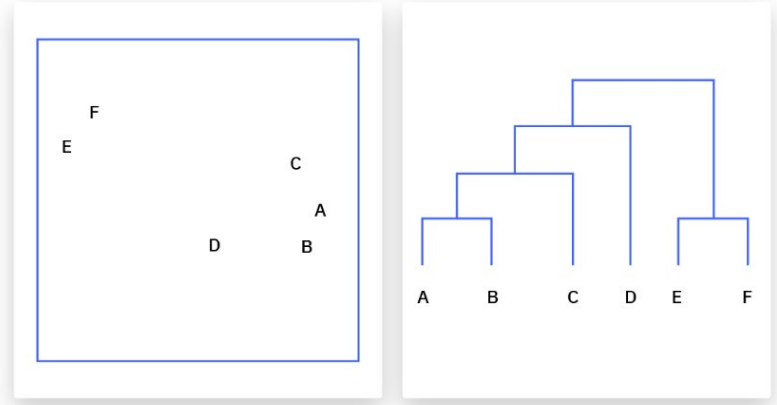
Credit: Analytics Vidhya

- K-Means is one of the **oldest** and **most widely** used unsupervised learning algorithms.
- **1956**: Hugo Steinhaus introduced the foundational concept of clustering in mathematics, laying the groundwork for partitioning data into groups.
- **1957**: Stuart Lloyd developed the k-means algorithm at Bell Labs, but his work remained unpublished until 1982.
- **1965–1967**: Various discrete and continuous versions appeared; Forgy and Jancey contributed to early implementations.
- **1967**: James MacQueen formally named and generalized the “k-means” method.
- **1970s -** : The method became a standard in cluster analysis.

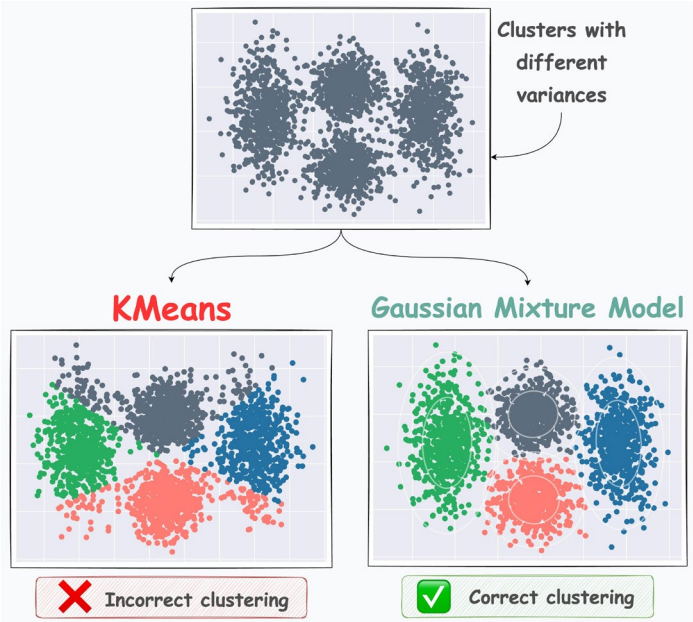
Other Major Clustering Algorithms

- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**: Identifies clusters of varying shapes and finds outliers using data point density. See demo here.
- **Hierarchical Clustering**: Produces a tree of clusters, available in agglomerative (bottom-up) and divisive (top-down) forms.
- **Gaussian Mixture Models (GMM)**: Model-based approach where each cluster is treated as a Gaussian distribution, allowing soft assignments of data points.

Recommended reference: "Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow" by Aurélien Géron, eBook available via UB Library.

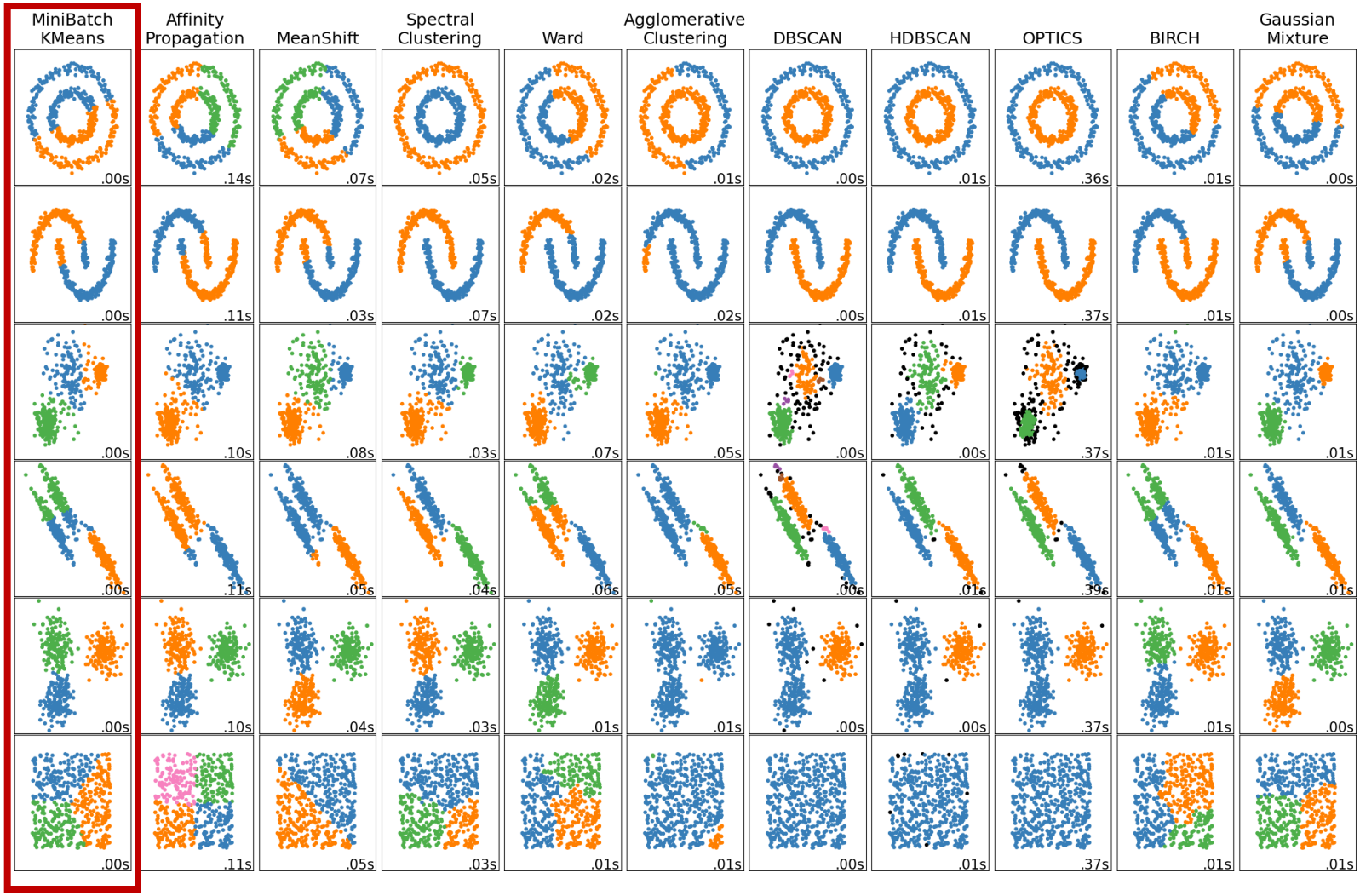


Credit: IBM



Credit: Daily dose of data science

Benchmark of Clustering Methods on Different Data Types

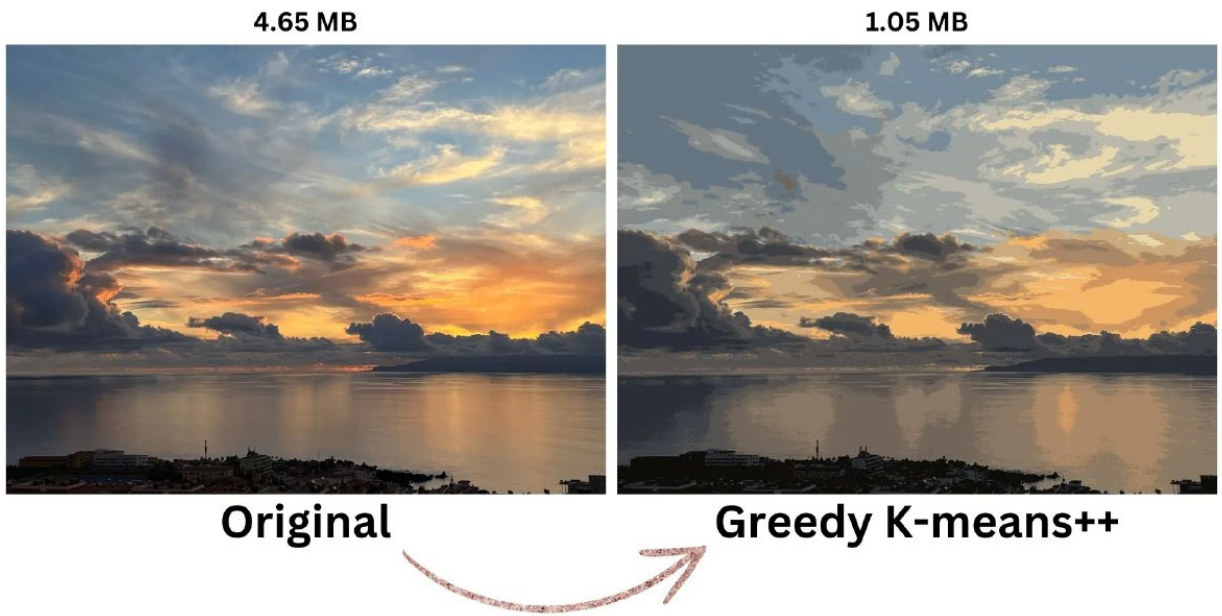


https://scikit-learn.org/stable/modules/clustering.html

Common Applications of Clustering Methods

Image Compression/segmentation

- Treat pixels as data points and cluster similar colors
- Reduces number of colors while preserving visual quality



Credit: rdiachenko.com

Iteration 1

Compressed Image

Credit: Medium

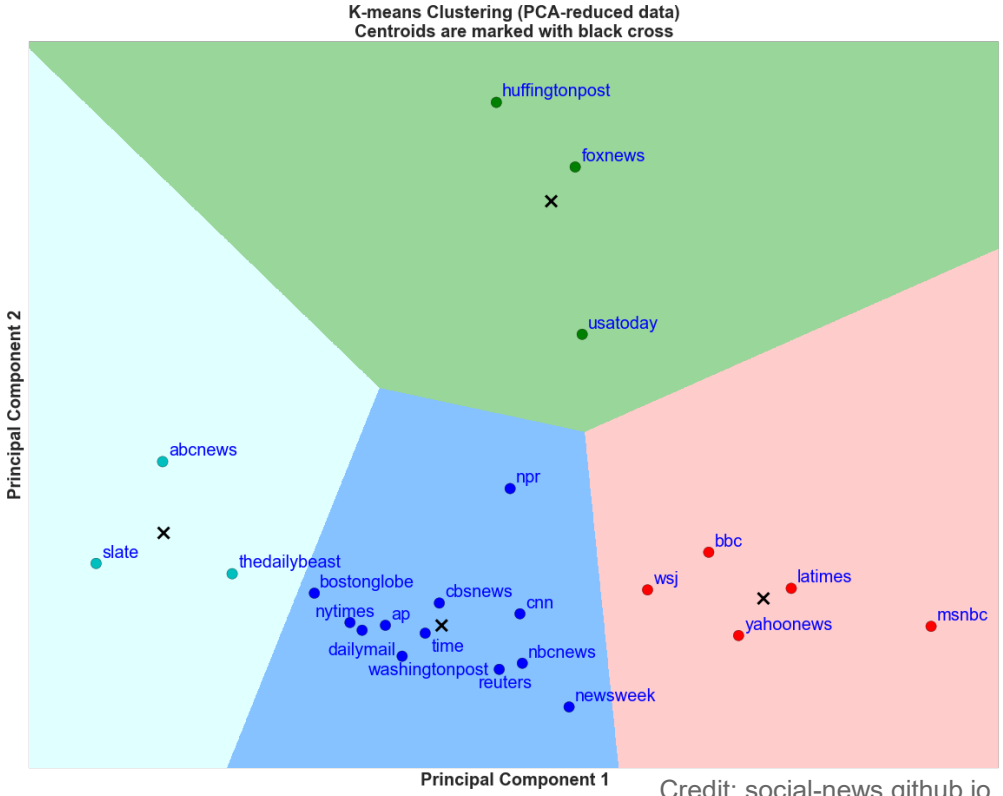
Common Applications of Clustering Methods (cont'd)

Social Media Analysis

- Cluster users by their social activity, preferences, and engagement
- Enables segment-based targeting and improved user experience

Recommendation Systems

- Cluster users by item preferences to infer recommendations



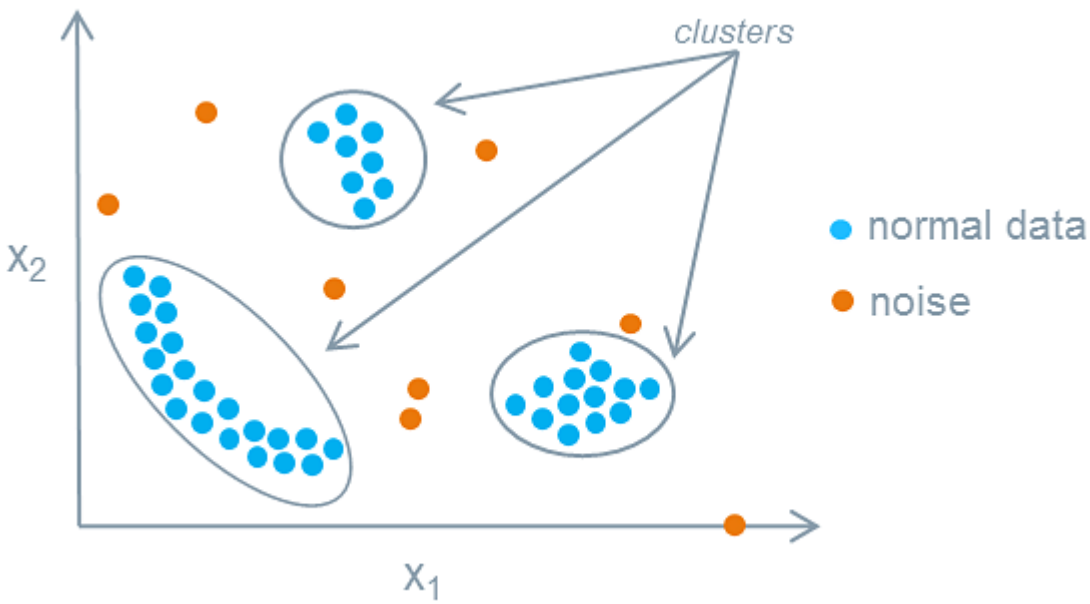
Credit: social-news.github.io

- **Green group:** High Facebook likes/shares per Bitly click and average Twitter engagement, indicating users often react to headlines without reading articles.
- **Red group**: High engagement on both Facebook and Twitter.
- **Teal group**: Low Twitter engagement, opposite trend of the red group.
- **Blue group**: Average Twitter engagement and average or below-average Facebook engagement.

Common Applications of Clustering Methods (cont'd)

Anomaly Detection

- Use clustering to define "normal" groups, then flag outliers
- Useful in fraud detection, network security, and quality control

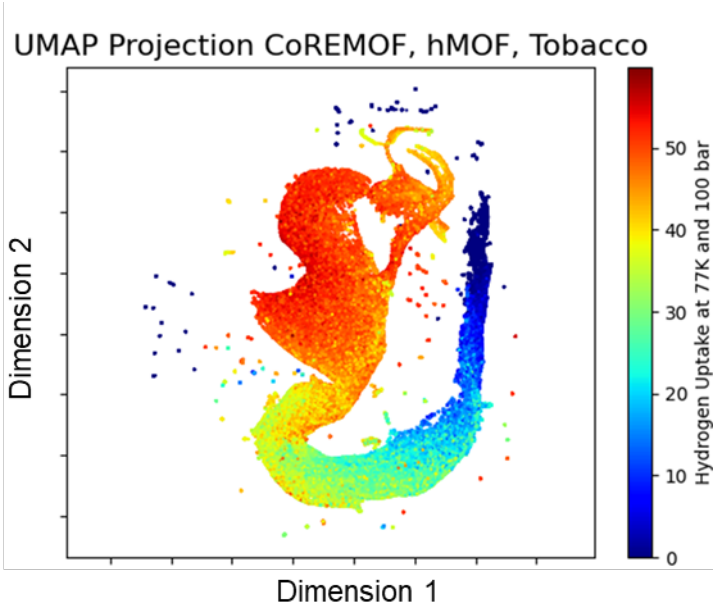


Credit: ProjectPro

Unsupervised Learning: Dimensionality Reduction

Another natural question arises: how can we see these clustered groups, especially when our data lives in very high dimensions? In fact, many real-world datasets—like spectra, images, or process variables—have dozens or even hundreds of features, making them difficult to interpret or visualize directly.

Dimensionality reduction techniques help us by **compressing the high-dimensional** data into **lower dimensions** while retaining as much of the essential structure as possible. This not only allows us to visualize complex datasets in two or three dimensions, but also often enhances clustering by filtering out noise and highlighting the most meaningful patterns.



UMAP visualization of metal-organic frameworks (MOFs) colored according to H_2 uptake, using 4 features: void fraction, volumetric surface area, pore limiting diameter, and largest cavity diameter (credit: Julia Merlin)

https://developers.google.com/machine-learning/intro-to-ml/what-is-ml

Uniform Manifold Approximation and Projection (UMAP)

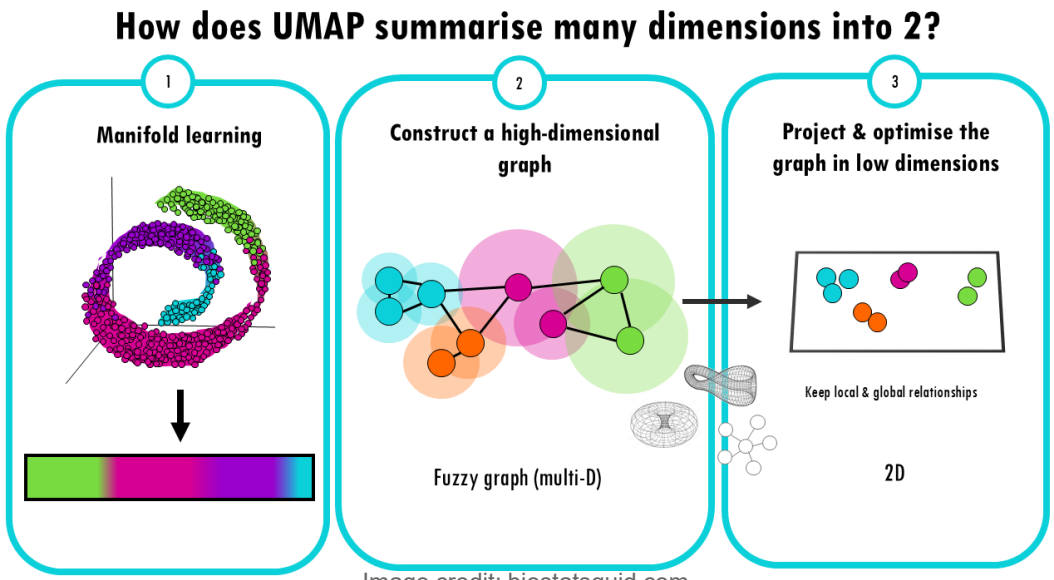


Image credit: biostatsquid.com

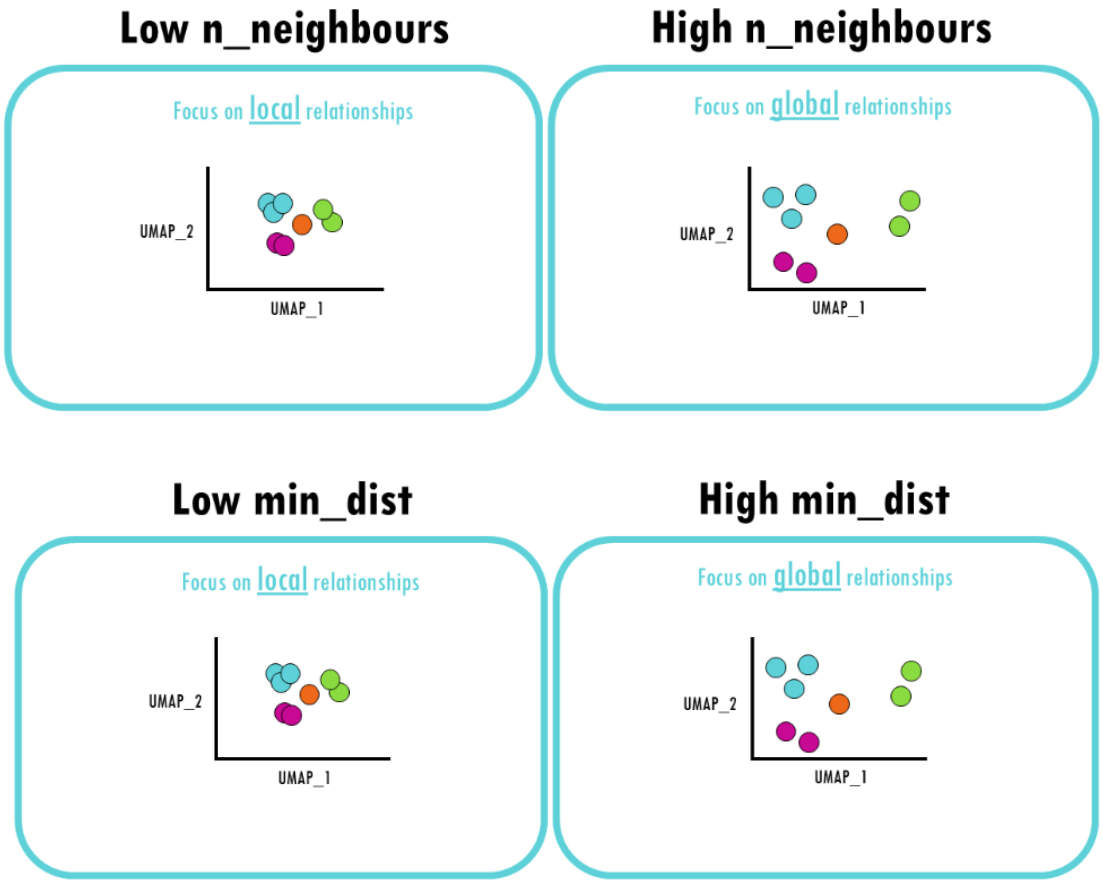
1. UMAP first **estimates this manifold structure** by looking at how points are connected to their neighbors.
2. UMAP builds a **high-dimensional fuzzy graph**; each data point is connected to its nearest neighbors, with edge weights showing how close or similar they are.
3. Once the high-dimensional graph is constructed, UMAP optimizes the layout of a low-dimensional analogue to be as similar as possible; there are hyperparameters to control how much local vs global structure is preserved.

[1] UMAP Uniform Manifold Approximation and Projection for Dimension Reduction | SciPy 2018 |
[2] L. McInnes, J. Healy, N. Saul, and L. Großberger, "UMAP: Uniform Manifold Approximation and Projection," J. Open Source Softw. 3(29), 861 (**2018**).

Uniform Manifold Approximation and Projection (UMAP)

n_neighbours is the number of nearest neighbors used to build the **initial high-dimensional graph**; it controls how much of the local vs global structure is preserved.

min_dist is the minimum allowed distance between points in the **low-dimensional space**.



Let's check out this demo!

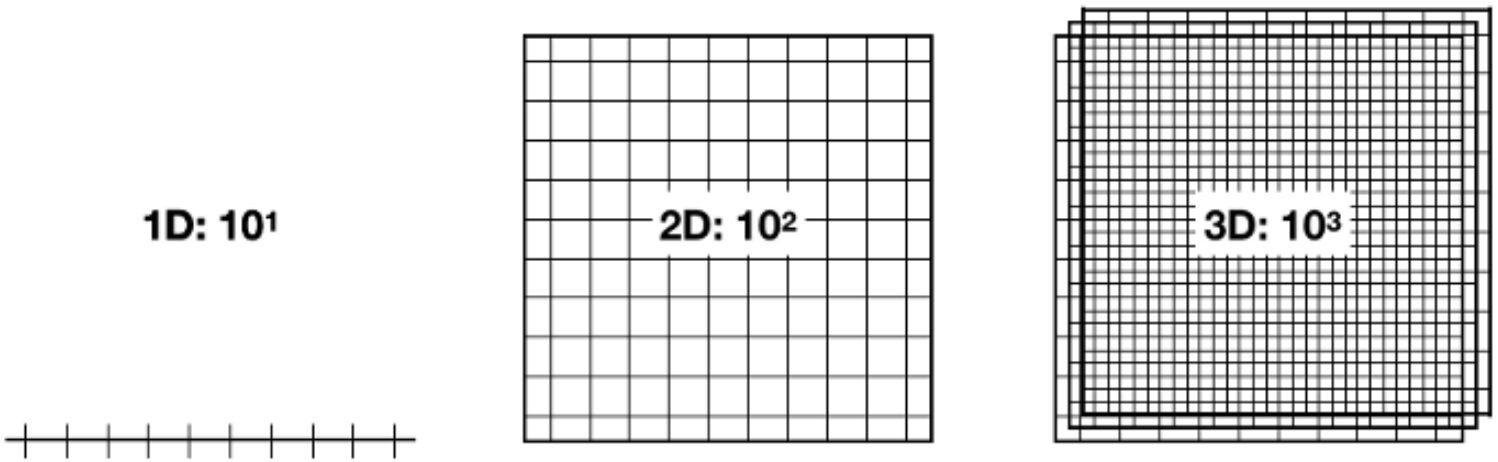
https://biostatsquid.com/umap-simply-explained/

Major Dimensionality Reduction Algorithms

- **Principal Component Analysis (PCA)**: A linear technique that transforms correlated variables into uncorrelated principal components maximizing variance.
- **Multidimensional Scaling (MDS):** A technique that reduces dimensionality by representing data points in a lower-dimensional space while preserving the pairwise distances or similarities between them.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE) [1]:** Non-linear technique good for visualization by preserving local neighbor structure in reduced dimensions.
- **Autoencoders:** Neural network-based methods that learn an efficient encoding for dimensionality reduction.

[1] https://distill.pub/2016/misread-tsne/

Curse of Dimensionality



The number of features required to keep average distance constant grows exponentially with the number of dimensions.

- The curse of dimensionality is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse.
- This sparsity is problematic for any method that requires statistical significance. In order to obtain a statistically sound and reliable result, the amount of data needed to support the result often grows exponentially with the dimensionality.
- Dimensionality reduction methods are helpful to overcome this problem.

https://en.wikipedia.org/wiki/Curse_of_dimensionality
https://builtin.com/data-science/curse-dimensionality

Clustering vs. Dimensionality Reduction

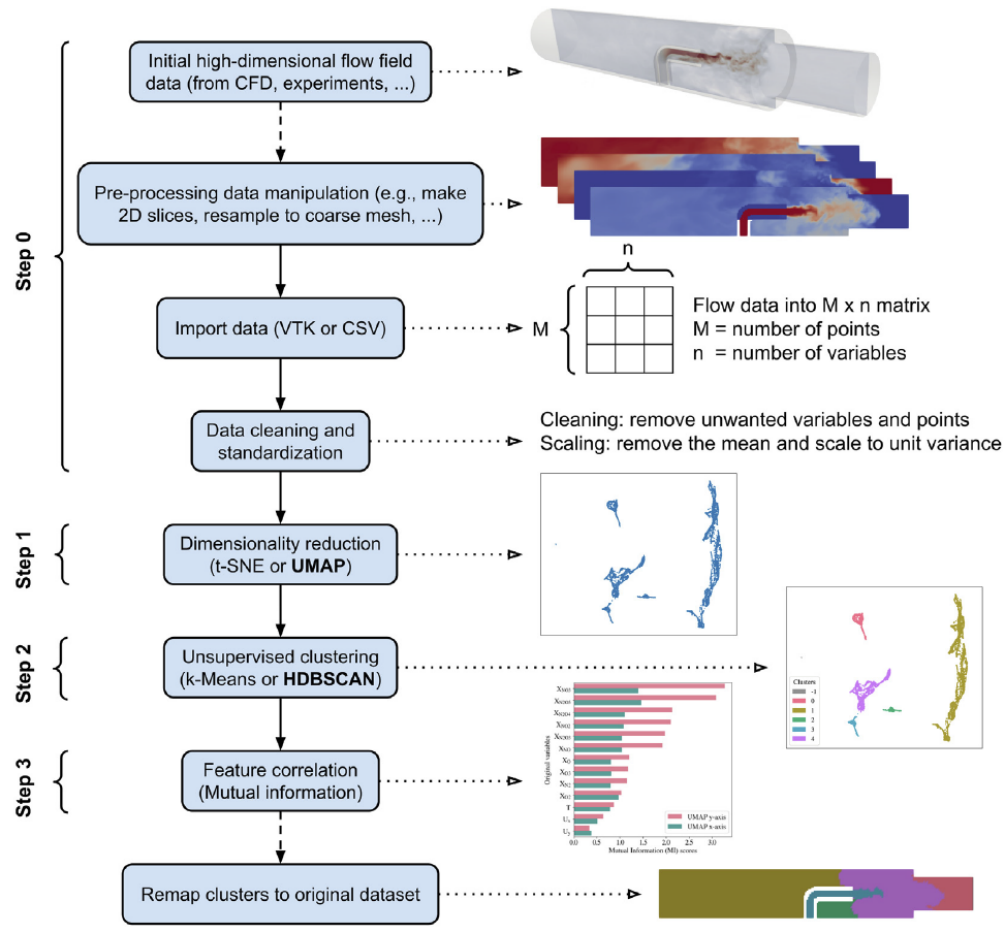
	Clustering	Dimensionality Reduction
Goal	Group similar data points into clusters	Represent data in fewer dimensions
Output	Cluster labels (which group each point belongs)	New lower-dimensional features/coordinates
Data Dimension	Keeps original feature space unchanged	Transforms data into a low-dimension representation
Focus	Finding structure (patterns, groups)	Simplifying structure (removing redundancy)
Use Case	Discover unknown categories	Visualization, noise reduction, preprocessing
Chemical Eng. Example	Grouping catalysts by performance profile	Reducing 50 process variables into 2–3 key axes

Applications of Unsupervised Learning in ChemE:
Reactive Flow Analysis

Background: Reactive-flow simulations generate massive multidimensional datasets (species mass fractions, temperature, velocity, etc. at every spatial point), making manual pattern recognition nearly impossible. Traditional analysis relies on expert intuition and 2-D slices that miss complex interactions.

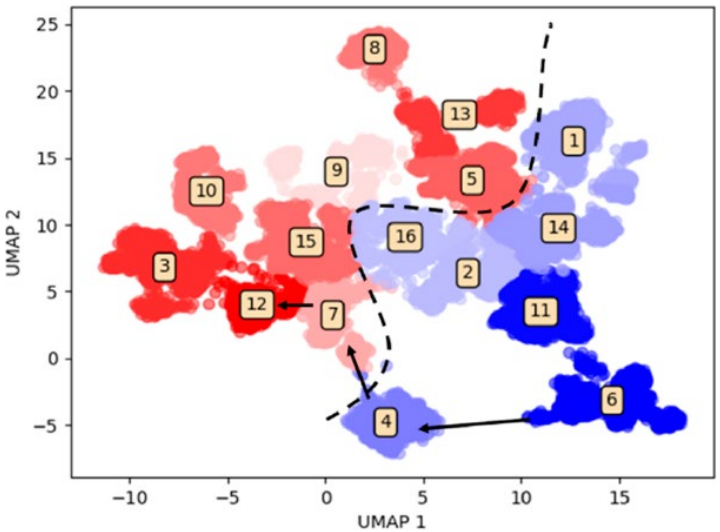
How UL was used: Applied **UMAP/t-SNE** to reduce high-dimensional simulation data to 2-D. Used **k-means** to automatically cluster regions with similar reaction and flow behavior. Correlated each cluster with key chemical/physical variables (mutual-information analysis).

Why clustering helps: Reveals hidden structure in complex reactive fields without prior labels. Converts simulation outputs into physically interpretable "**zones**." Enables faster insight and reduces subjective bias in post-processing.

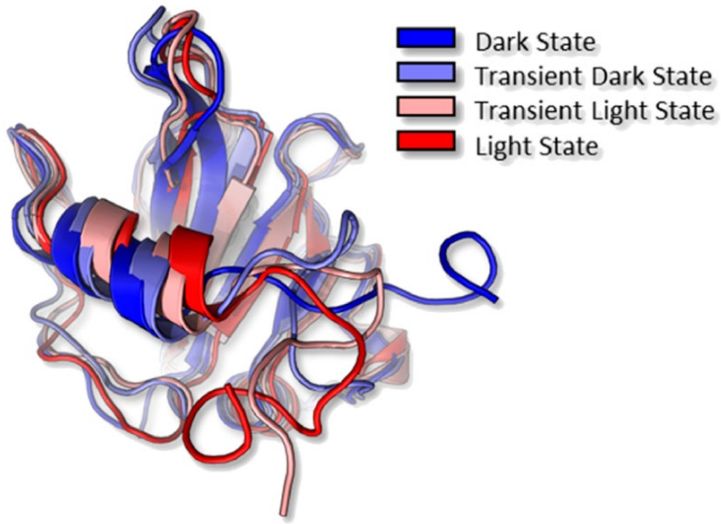


Rovira, Marc, Klas Engvall, and Christophe Duwig. "Identifying key features in reactive flows: A tutorial on combining dimensionality reduction, unsupervised clustering, and feature correlation." *Chemical Engineering Journal* 438 (2022): 135250.

Applications of Unsupervised Learning in ChemE:
Clustering and Dimensionality Reduction in Biomolecular Dynamics



UMAP 2D projection with the reduced space clustered in 16 macrostates/conformations



Visualization of the four representative states involved in the allosteric process

Background: Protein dynamics determine biological function, but molecular dynamics (MD) simulations generate extremely high-dimensional data—millions of atomic coordinates over time. To understand conformational transitions and allosteric mechanisms, researchers need to project this data into a lower-dimensional space that preserves both structure and kinetics.

How clustering was used: Applied **UMAP** to embed 12 μs of MD data into 2 dimensions that preserve both structural and kinetic relationships. Performed **k-means clustering** on the UMAP projection to group conformations into 16 macrostates, each representing a metastable basin on the protein's free-energy surface.

Trozzi, Francesco, Xinlei Wang, and Peng Tao. "UMAP as a dimensionality reduction tool for molecular dynamics simulations of biomacromolecules: a comparison study." *The Journal of Physical Chemistry B* 125.19 (2021): 5022-5034.